Health Copilot

Your medical papers, read, checked and explained, in your language

Prescriptions · lab reports · discharge summaries → one verified, ABDM-ready health record

Team hack_masters · Altrix Labs Hackathon · AI-Powered Personal Health Copilot

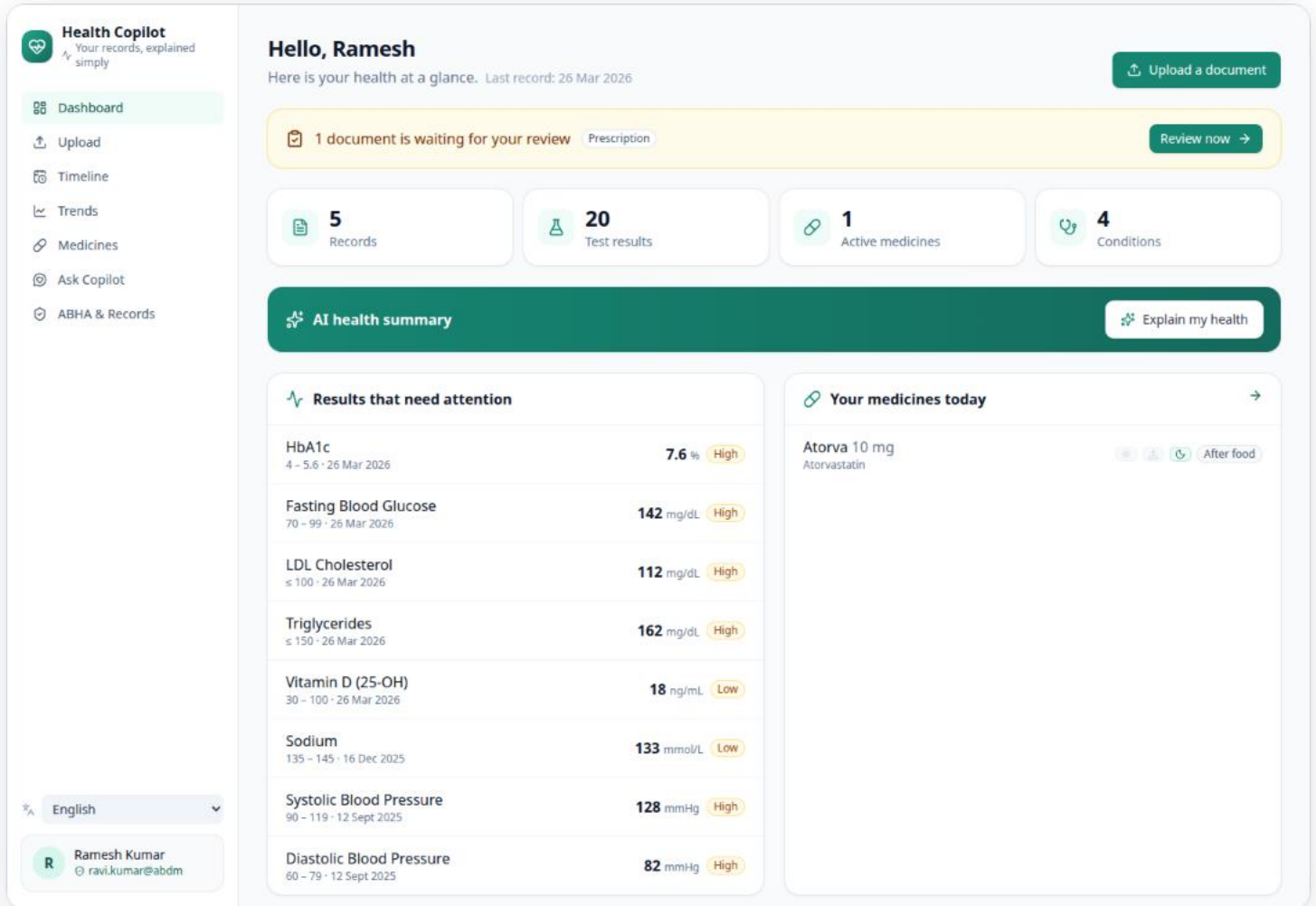
The problem

- Indian patients carry their history on **paper**: handwritten prescriptions, lab printouts in `lakhs/cumm` and `gm%`, and multi-page discharge summaries.
- The documents mix **English with Hindi, Telugu, Tamil or Kannada**.
- Nobody explains the abnormal values. Old prescriptions linger. Records don't move between hospitals.
- ABDM can carry records between providers, but only if they are **structured FHIR** linked to an **ABHA**.

What a patient actually needs

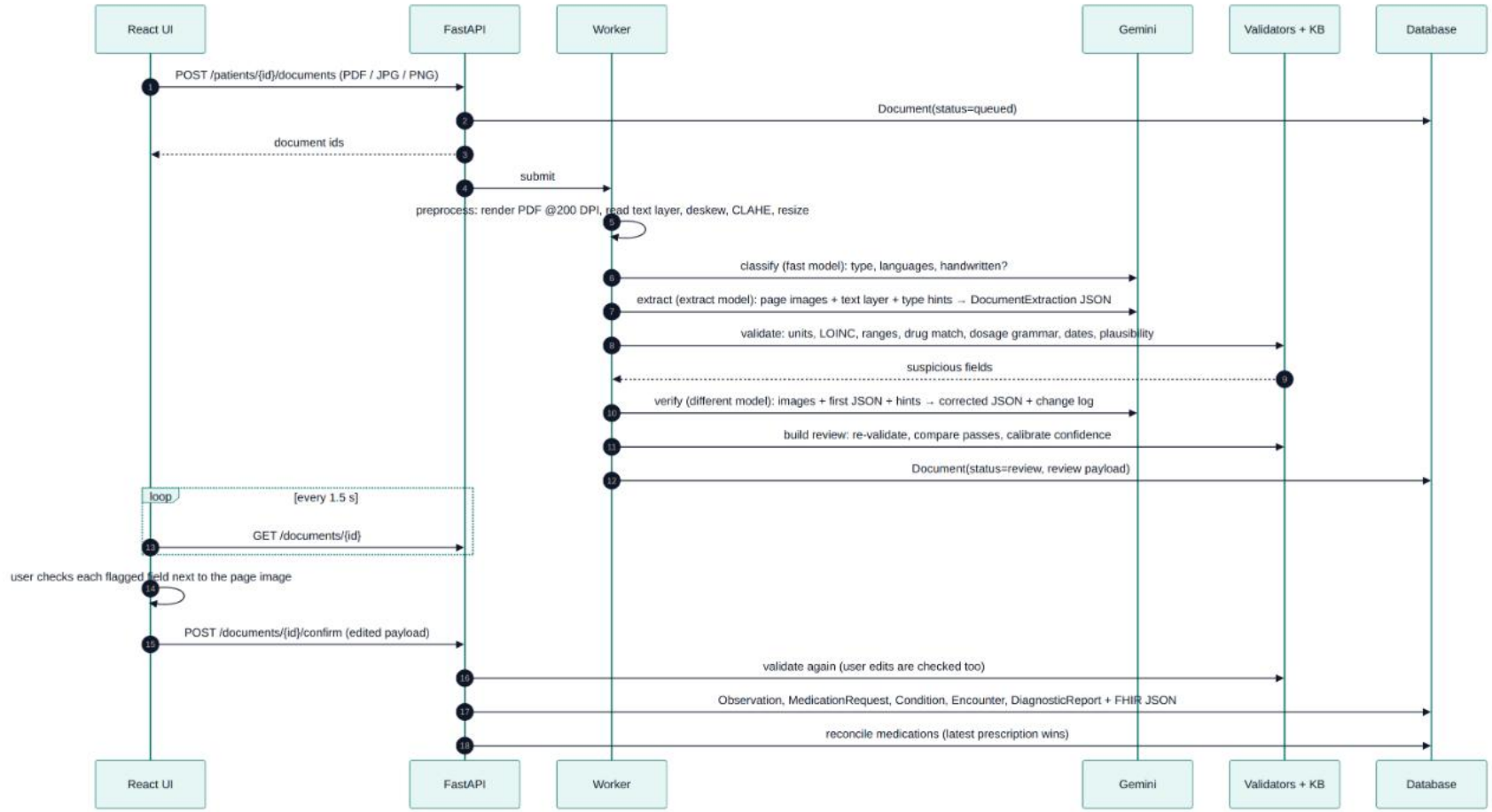
1. Snap a photo and get the data out *correctly*.
2. "What does this result mean? What do I ask my doctor?"
3. One timeline: medicines, results and conditions.
4. In their language.
5. Share it through ABHA.

What we built



Upload or photograph → AI extraction with a second-model check → side-by-side review → unified profile, timeline, trends, medicine schedule, grounded summaries and chat → FHIR / ABHA

AI pipeline: the LLM reads, code decides



Gemini vision with schema-constrained JSON · a **different model** re-reads the page, guided by the validator's suspicious fields · deterministic checks own units, ranges, flags, drug codes, dosage grammar and dates · a human confirms before anything reaches the record

What the code guarantees

Deterministic validation

- 60 LOINC tests with unit conversion and sex-specific ranges. The report's own range is used first.
- High, Low and Critical flags are **computed, not generated**.
- 213 Indian brands mapped to generics. Combination brands stay distinct: *Glycomet GP ≠ Glycomet*.
- Dosage grammar: `1-0-1`, `BD`, `HS`, `SOS`, `AC/PC`, weekly, `x 1 month`.
- Day-first Indian dates, BP split into systolic and diastolic, impossible values blocked.

Grounded explanations

- Summaries use a fact sheet built from the DB. The model writes prose only.
- Values, flags, schedules and critical banners are attached by code.
- A safety filter blocks dosing advice and diagnoses, and checks that every number is grounded.
- Translations must preserve every number.
- The chat answers only from the patient's records, with citations, and shows an emergency notice for red-flag symptoms.

Measured, not claimed

On 10 documents / 96 items	Single pass	+ Verify
Field accuracy, end to end	99.8%	**100%**
Medicines F1	100%	100%
Lab / vital results F1	99.2%	**100%**
Diagnoses F1	100%	100%

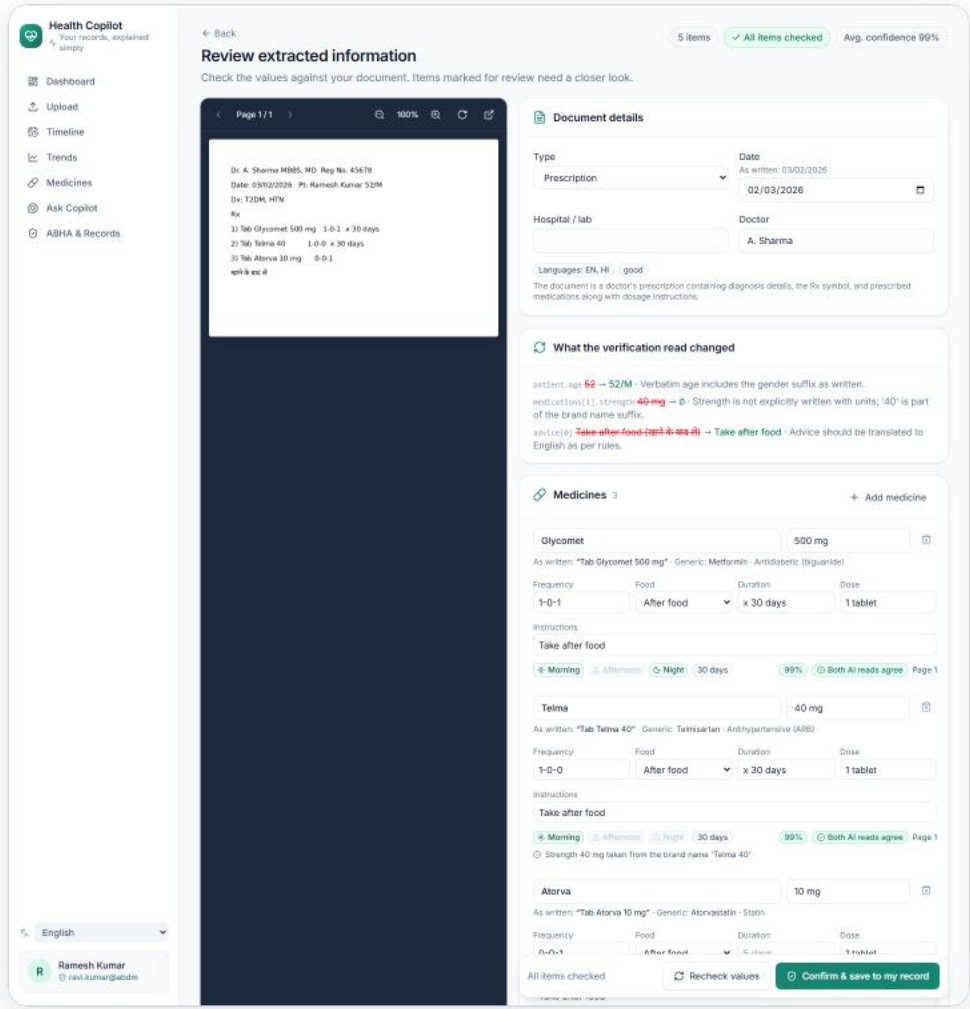
Test set: printed, handwritten, phone-photo, multi-page PDF, Hindi, Telugu, Tamil and Kannada documents. Hand-written ground truth; `eval/run_eval.py`.

What the evaluation caught (and we fixed)

- The verifier recovered the one value the first pass missed: SpO2 in narrative text.
- Rewording between the passes caused false review flags. We now match on LOINC, generic and ICD-10 codes, so false alarms drop from about 20% to about 1%.
- Combination brands were being collapsed.
- Older prescriptions were staying active.

The samples are synthetic, so real handwriting will be harder. That is why every document is reviewed.

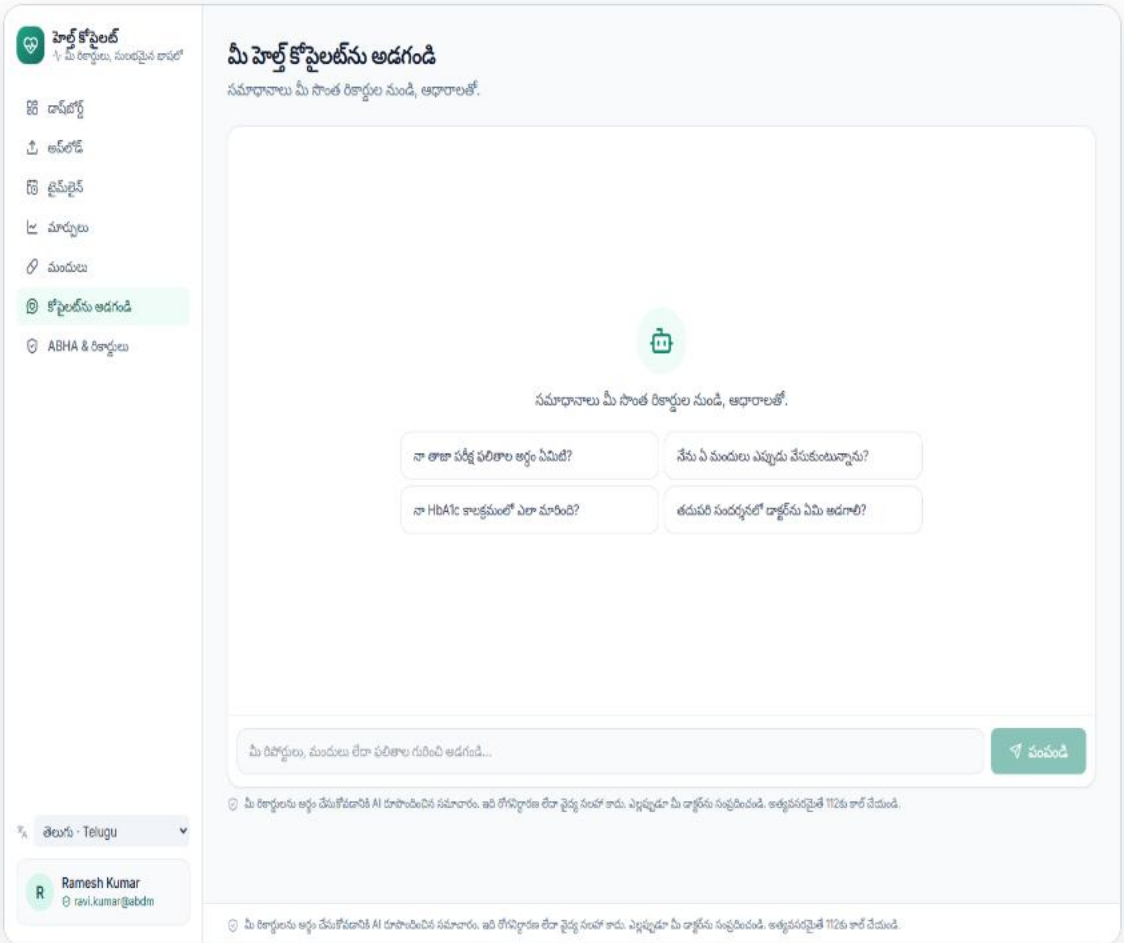
Review UX: human in the loop, where it matters



The deskewed page sits next to editable fields. Badges show confidence, verifier changes and validation issues. Only flagged items need attention; "Looks right" accepts in one tap.

In your language



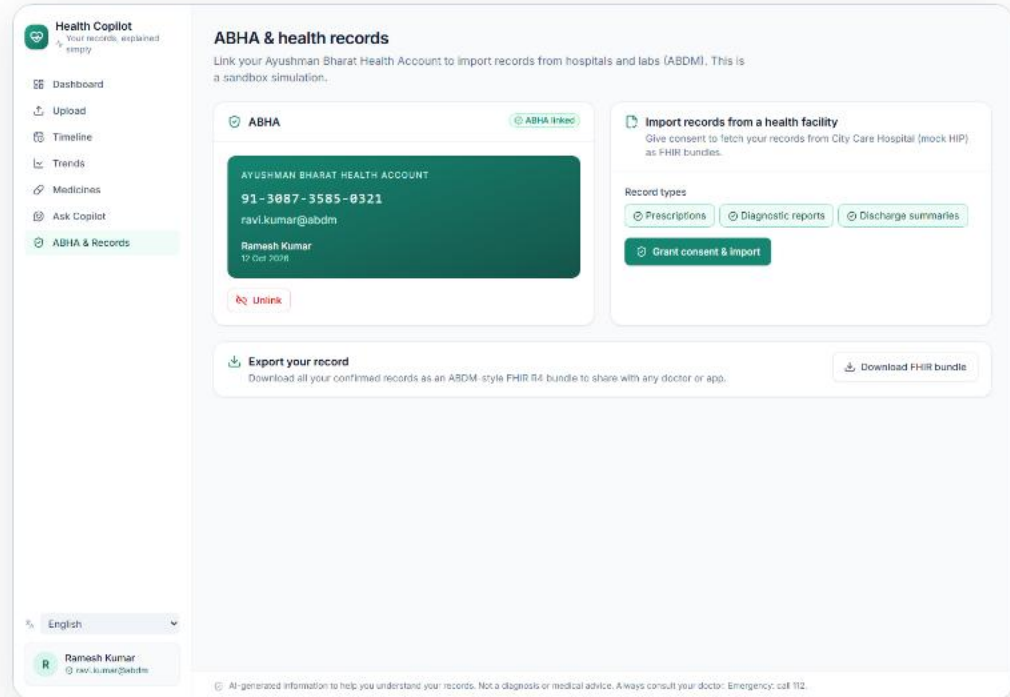


The UI and AI summaries are available in English, हिन्दी, తెలుగు, தமிழ் and ಕನ್ನಡ. Medical terms keep the English term in brackets.

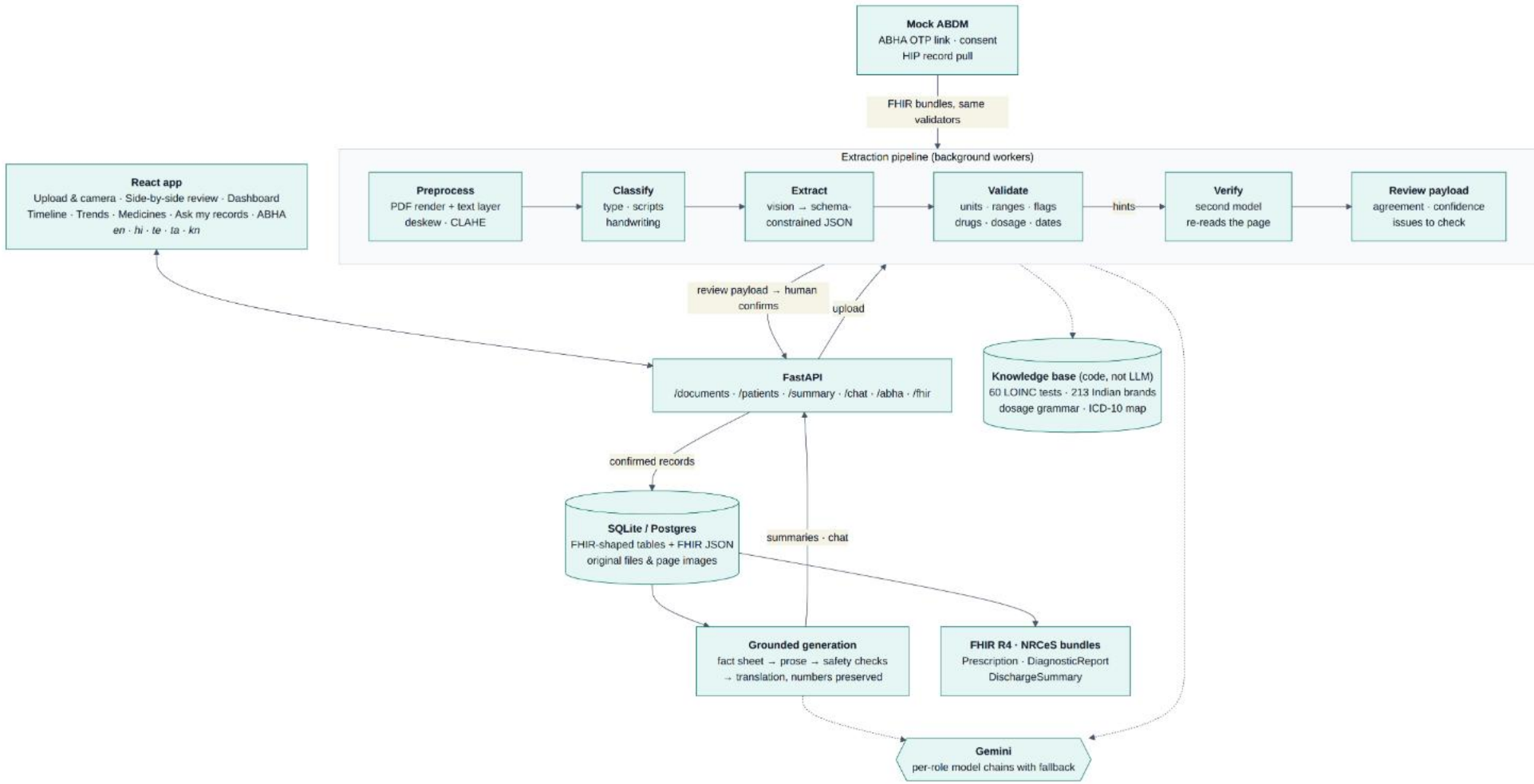
Bilingual and regional-script documents are extracted directly.

ABDM-ready

- Every confirmed document becomes an **NRCeS-profile FHIR R4 Bundle**: PrescriptionRecord, DiagnosticReportRecord or DischargeSummaryRecord. Resources are validated with `fhir.resources`.
- Codes: **LOINC, UCUM, ICD-10, SNOMED** composition types and v3 interpretation codes.
- The **ABHA** number and address are Patient identifiers. The whole record exports as one bundle.
- **Mock ABDM flow:** OTP link, consent artefact, and a pull from a Health Information Provider (HIP). Imported bundles go through the **same validators**.
- Going live means swapping one module for the sandbox gateway.



Architecture



FastAPI · SQLModel (SQLite or Postgres) · background workers · Gemini model chains with fallback (OpenAI drop-in) · React 19 + Vite + Tailwind + TanStack Query + Recharts + i18next · `docker-compose up`

Impact and next steps

Who it helps

- Chronic patients (diabetes, BP, thyroid) tracking trends across labs and hospitals.
- Caregivers managing a parent's medicines and schedules.
- Doctors receiving a structured, ABHA-linked history instead of a paper folder.

Next steps

- ABDM sandbox: HIU registration and Fidelius encryption.
- Medicine reminders from the extracted schedule, plus interaction checks.
- Grow the evaluation set with real, consented documents, and calibrate confidence on it.
- Voice summaries in regional languages.

> AI reads and explains. Code checks. You confirm. Your record, in your language, ready for ABDM.